NASHER MUSEUM · PHOTOGRAMMETRY COLLECTION

# Artifact Viewer Handbook

How to open the artifact viewer, keep it on your computer like any other app, and put your own scanned objects into it. No software to buy, nothing to set up.

## Open it

The viewer is a web page. You do not download or install anything to use it — you visit an address, the same way you would visit any website. Here it is:

THE ADDRESS

**nikkinaq-a11y.github.io/artifact-viewer**

Works on any reasonably modern Mac or Windows computer, in Chrome, Edge, Safari or Firefox. Nothing is uploaded and no account is needed.

You will see an empty pedestal in a dark room. **That is correct.** The viewer arrives empty on purpose, because the museum's scans are not published on the open internet. You add objects yourself, or open a collection someone sends you — both are covered below.

> **Where the address comes from.** The viewer is stored on GitHub, a website developers use to keep their work. The GitHub page at github.com/nikkinaq-a11y/artifact-viewer holds the instructions a computer follows to build the viewer; the address above is the finished, usable version. If someone sends you the GitHub page and you only see folders and files, you are in the workshop rather than the gallery — use the address above instead.

## Two ways to use it

Both run exactly the same viewer. The difference is whether it lives in a browser tab or behaves like an installed program.

| NOTHING TO INSTALL | RECOMMENDED IF IT IS YOUR COMPUTER |
| --- | --- |
| **In a browser tab**<br>Visit the address and use it. Best for showing someone quickly, or on a computer that is not yours.<br>Needs an internet connection the first time. | **Installed as an app**<br>Sits in your Dock or Start menu with its own icon, opens in its own window with no address bar, and works with no internet at all.<br>Takes about ten seconds to set up, once. |

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## Install it as an app

This does not download an installer or ask for a password. The browser keeps a copy of the viewer on your computer and gives it an icon.

### On Mac or Windows, using Chrome or Edge

01 Open the address in Chrome or Edge.

02 Look at the right-hand end of the address bar for a small install icon — a screen with a downward arrow.

> If you cannot see it: open the browser's three-dot menu and look for *Cast, save and share* → *Install page as app*, or simply *Install*.

03 Click it, then click **Install**.

04 The viewer opens in its own window and now appears in your Applications folder or Start menu. Open it from there in future.

### On a Mac, using Safari

01 Open the address in Safari.

02 From the menu bar choose **File → Add to Dock**.

03 Confirm the name. It now sits in your Dock like any other app.

> **It genuinely works offline.** Once installed, the viewer opens on a plane, in a storage room, or in a gallery with no guest wi-fi. Your objects are kept on your own computer, not on a server.

### Or download the Mac app

A proper Mac application is also available, for anyone who would rather have an ordinary program than a browser install. It is the same viewer, and it works with no internet at all.

01 Download it from this link. It is also linked at the bottom of the on-screen controls (press `Tab`).

   The link always gives the newest version.

02 Open the downloaded file and drag **Artifact Viewer** into Applications.

03 The first time you open it, macOS will refuse, because it did not come from the App Store. Open **System Settings → Privacy & Security**, scroll down, click **Open Anyway** next to Artifact Viewer, and confirm. After that it opens normally.

In the Mac app, saving a snapshot or exporting a gallery asks where to put the file, like any other program, starting in your Downloads folder.

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## Put an object on the pedestal

**Drag a 3D file from your desktop onto the viewer window and let go.** That is the whole procedure. The object is measured, sized to the pedestal and lit automatically. It stays in your collection until you remove it, including after you close and reopen the viewer.

If you would rather pick from a list, click the **Gallery** tab in the top-right corner, then **+ Add artifact**.

### File types it accepts

| TYPE | WHAT TO EXPECT |
|---|---|
| `.glb` `.gltf` | The most reliable. Everything the object needs is inside one file. |
| `.fbx` | Usually fine. Some carry their colour image separately — see the grey-object note below. |
| `.obj` | Drag the `.obj`, its `.mtl` and its image files in together, all selected at once. |
| `.ply` | Common from scanners. Colour is usually built into the surface. |
| `.stl` | Shape only. This format cannot store colour, so it always looks plain. |
| `.usdz` `.usd` | Supported, but less thoroughly tested than the rest. |

> **If an object arrives grey or black.** Nothing is broken. Some files record only the *location* of their colour image on the computer that made them — a path your computer cannot follow. Select the 3D file *and* its `.jpg` together and drag them in as one selection. The colour appears, and it is remembered from then on.

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## Send someone a collection

Your objects live on your computer only. To let a colleague see the same set, you send them a collection file.

01 Open the **Gallery** tab and click **Export gallery**. You get a single `.zip` file.

It contains every object, its colour images, and the way you had the scene arranged — pedestal size, lighting, background, camera.

02 Send them that file, along with the address at the top of this page.

Five masks come to roughly 25 MB, which emails comfortably. Very large scans will need a shared drive instead.

03 They open the viewer, click **Gallery → Import gallery**, and choose the file. They see exactly what you staged.

> **Importing replaces what is already there.** A collection arrives as the curated set it was meant to be, rather than mixing into someone's existing objects. Export your own collection first if you want to keep it.

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## Moving around

**Click and drag** to walk around the object. **Scroll** to move closer or further away. Everything else is a single key — press it, and press it again with `Shift` held to go the other way.

| KEY | WHAT IT DOES |
|---|---|
| `C` | **C**amera — next of five fixed views, all from the front: straight on, from above, from below, close and wide (turn the object with R to see its other sides) |
| `D` | **D**istance — far, then mid, then near, then back to far |
| `O` | Next **o**bject in your collection |
| `R` | **R**otate the object 30° on its base |
| `S` | **S**cale — make the object larger |
| `H` | **H**eight — raise the pedestal, and the object with it |
| `W` | Pedestal **w**ider |
| `Z` | Pedestal deeper, front to back |

| KEY | WHAT IT DOES |
| --- | --- |
| `P` | **P**edestal — standard, then hidden (the object floats), then white, then back to standard |
| `L` | **L**ighting — switch between light and stark. Use the light slider on the right edge to make either one brighter or dimmer. No effect in silhouette, where the whole rig is off |
| `B` | **B**ackground scene — dark room, then white, then horizon, then silhouette |
| `I` | Save an **i**mage — a picture of the scene, without any of the on-screen panels |
| `Tab` | Show this list on screen. The readout in the corner collapses too — click the camera chip at the lower left |

The four scenes suit different jobs: **dark** is the gallery look, **white** is for documentation photographs, and **horizon** puts a pale blue floor against a grey sky, so you can see exactly where the object is standing. Its horizon sits at true eye level, so it moves as you raise and lower the camera, just as a real horizon does. There are no walls in any scene; in dark and white the floor fades into the background.

**Silhouette** is not simply a fourth background. The floor disappears, the background turns pure white and every lamp in the room goes out, so nothing falls on the object at all and it reads as a flat black cut-out against white — an outline for comparing profiles, not a surface for reading detail. Because the rig is off, `L` does nothing here, and the on-screen readout says so.

`D` gives more framings rather than more angles. It holds whichever view you are on and starts far back, then moves in twice, each step a good deal closer, keeping the lens the same, so each reads as the same shot from nearer rather than a zoomed one.

The two lighting settings on `L`: **light** is the everyday setting, shaped light from the upper left with the shadow side still readable. **Stark** is one hard light from the side and a rim from behind, with nothing filling in, so half the object falls away into darkness. The **light** slider on the right edge has five steps, from dim

to bright, and works on either setting. Pale objects usually look best on a lower step.

On the **from above** and **from below** views, a **camera height** slider appears at the right edge. It raises and lowers the camera without changing the angle it looks at the object from. Each of the two views remembers its own setting, and double-clicking the slider resets it.

`I` saves a snapshot as a PNG image. It is named after the object, the view and the distance, so a set of snapshots sorts itself.

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## When something looks wrong

**The pedestal is empty and nothing happens**

The viewer starts empty by design. Drag a 3D file onto the window, or import a collection file from a colleague.

**My objects have vanished**

Objects are stored by the browser, so clearing your browsing data or history removes them, and they do not follow you to another computer or another browser. Keep your original files, and export a collection file for anything you would hate to rebuild.

**A large scan is slow, or the tab freezes**

Very large scans — over roughly 100 MB — are demanding. Close other tabs and try again. If a particular file always fails, it likely needs to be reduced in size before it can be used comfortably.

**The object is on its side or far too small**

The viewer sizes and seats each object automatically, but it cannot know which way up a scan was saved. Use `R` to turn it and `S` to resize.

**I opened the GitHub page and it is full of files**

That is the workshop, not the gallery. Use the address at the top of this page.

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Viewer: nikkinaq-a11y.github.io/artifact-viewer

Source: github.com/nikkinaq-a11y/artifact-viewer

Rebuilt from the Unreal Engine `Artifact_Setup` viewer, so artifacts can be added by dragging in a file.